Ahmed Mishqat

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RESEARCH PROFILE

Second-year BEng Mechanical Engineering student at the University of Manchester (First-Class Honours, Bicentenary Global Futures Scholar) with a strong foundation in materials science, structural mechanics, and experimental data analysis. Particular academic interest in microstructure–property relationships, deformation mechanisms, and advanced manufacturing processes, developed through coursework in Materials 2 and hands-on lab and simulation work. Seeking research experience to deepen this foundation and explore a potential pathway to postgraduate study.

EDUCATION

The University of Manchester — BEng Mechanical Engineering *2024 – 2026 (Expected)*

- **Classification:** First-Class Honours (First Year)
- **Scholarship:** Bicentenary Global Futures Scholarship — the University's highest merit-based international award.
- **Relevant Modules:** Materials 2 (microstructure, phase diagrams, deformation mechanisms, creep, weld metallurgy, stainless steels), Thermodynamics, Fluid Mechanics, Structures, Manufacturing Engineering, Mechanical Engineering Systems

Mastermind English Medium School — CAIE A Levels *2022 – 2024*

- **Grades:** Physics (A*), Chemistry (A*), Mathematics (A*)
- **Awards:** Daily Star Award for Outstanding Academic Achievement; Intra-School Olympiad and Sports Awards

TECHNICAL SKILLS

- **Materials & Characterisation:** Microstructure–property relationships, deformation and fracture mechanisms, creep, weld metallurgy, sensitisation, phase transformations in Al alloys and steels; DBTT analysis and impact energy data modelling
- **Structural FEA:** SolidWorks Simulation (stress & displacement studies, mesh convergence); Abaqus (nonlinear & contact analysis)
- **CFD & Simulation:** Siemens Star-CCM+ — turbulence modelling, parametric mesh studies, post-processing
- **CAD & Design:** SolidWorks, Fusion 360 — Part & Assembly Modelling, Engineering Drawings, Design-for-Manufacture, Topological Optimisation
- **Workshop & Manufacturing:** Lathe & Milling, MIG/TIG Welding, 3D Printing, Bench Work, Assembly & Tolerancing
- **Data Analysis & Programming:** Python (NumPy, SciPy, Matplotlib), experimental data processing, curve fitting, Monte Carlo uncertainty estimation, FFT-based signal analysis, Arduino (embedded control, sensor fusion)
- **Documentation:** Technical report writing, engineering drawing production, research presentation; personal portfolio at mishqat.github.io

RELEVANT ACADEMIC & PROJECT WORK

Ductile-to-Brittle Transition Temperature Analysis *2025–2026*

Coursework — Materials 2

- Processed Charpy impact energy vs. temperature datasets to extract DBTT and upper-shelf energy, fitting a four-parameter hyperbolic tangent model using SciPy curve optimisation.
- Implemented a Monte Carlo uncertainty framework based on ASTM E23 error models to produce statistically rigorous confidence intervals on DBTT and USE — results presented in annotated multi-panel Matplotlib figures.
- Deepened understanding of notch toughness, fracture mechanics, and how microstructural variables (grain size, inclusion density, thermomechanical treatment) shift transition behaviour in structural steels.

IMechE Design Challenge 2026 — 1st Place, NW Regional Heat *Spring 2026*

Autonomous Pipe-Climbing Robot — Advancing to National Finals

- **Topological Optimisation & FEA:** Proposed and executed topology optimisation on primary structural components in SolidWorks, reducing part mass by over 30% — validated through iterative FEA stress and displacement studies with mesh convergence checks.
- **Materials Selection:** Informed material and manufacturing choices for structural and clamping components, balancing stiffness, weight, and fatigue life under cyclic loading conditions.

- **Systems Design:** Designed the main chassis and electronics enclosure; performed analytical torque and load calculations for motor selection across the full range of pipe diameters in scope.
- **Control:** Authored Arduino control architecture including a dead-reckoning algorithm using motor pulse timing when ultrasonic sensing proved unreliable at range.

IMechE CADathon 2026 — 2nd Place, Sustainability Category *Spring 2026*

CrushIt: Gamified Pedal Can Crusher — SDG 11 · 8-Hour Design Sprint

- **Materials & Manufacturability:** Specified materials (304 stainless steel platforms, HDPE injection-moulded housing, galvanised mild steel linkage, PP liquid tray) with written justifications covering corrosion resistance, fatigue life, food-grade compliance, and end-of-life recyclability.
- **Mechanical Design:** Engineered a pedal-actuated toggle linkage, carbonation-safe pre-puncture spike platform, liquid drainage system, and mechanical safety interlock — fully passive, zero-electricity operation.
- **Concept & Delivery:** Originated the core incentivisation concept and led SolidWorks CAD and full presentation within the 8-hour window; 2nd place awarded in sustainability design category.

Robotic Arm Poker Dealer, RoboSoc UoM *Spring 2025*

- Designed SolidWorks structural components for a servo-driven robotic arm; identified and documented root cause of actuation failure (actuator-mass mismatch), reinforcing systematic failure analysis skills.

LEADERSHIP & EXTRA-CURRICULAR

President — Mathematics Club, Mastermind English Medium School *2023 – 2024*

- Planned and directed Highschool Math-Fest, an annual competition with 70+ volunteers and 1,000+ participants; managed operations, logistics, and real-time problem-solving under pressure.